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Homework 4: Diffractive Optical Elements and Deconvolution

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The assignment report is due on 7 October, 2026. In it, you must solve the exercises presented in the guide in report format, and attach the files you generate, such as .py, ipynb, .mat, .m, .fig, etc. The report should briefly describe the experiments performed, including the following sections: summary, description of experiments, and conclusions. It is suggested that you prepare the report in LaTeX to facilitate writing equations. Send a compressed folder with the files to the Aula submission corresponding unit: , with the subject line: **Homework4_FotografiaComputacional** and the compressed folder's name **homework4_nameSurname.zip**, where Name is your first name and Surname is your last name. Caution: any attempt at copying or fraud will invalidate the lab and result in a zero on the lab grade.

1. Learning Goals

To identify the components of a compressive color diffractive system and solving the deconvolution task. Once the homework will be completed the student will be able to simulate a diffractive color optical system and solve the deconvolution task.

2. Introduction

Computational photography involves to study optical system that exploits diffractive optical elements (DOE) to encodes the incident field.

2.1. Forward Model

The compressive diffractive color measurement can be represented as:

$$\mathbf{y} = \text{Sensor}(\mathbf{h}_l * \mathbf{f}_l)_{\text{Bayer}} \quad (1)$$

where $\mathbf{y} \in \mathbf{R}^{N \times N}$. $\mathbf{f}_l \in \mathbf{R}^{N \times N}$ denotes the l^{th} incident field, $\mathbf{h}_l \in \mathbf{R}^{N \times N}$ denotes the point spread function (PSF), and $*$ is the convolution operation.

2.2. Deconvolution: Wiener Filter

The deconvolution process involves solving the following regularization problem

$$\min_{\tilde{\mathbf{f}}} \{ \|\mathbf{y} - \text{Sensor}(\tilde{\mathbf{f}}_l * \mathbf{h}_l)_{\text{Bayer}}\|_2^2 + \tau \|\tilde{\mathbf{f}}_l\|_2^2 \} \quad (2)$$

τ is a regularization parameter.

$$\hat{\mathbf{y}}_l = \mathcal{F}^{-1} \left\{ \frac{\mathbf{H}_l^*}{\mathbf{H}_l^2 + \tau} \mathbf{Y} \right\} \quad (3)$$

where $\mathbf{y}$ correspond to the measurement in the Fourier domain, $\mathbf{H}_l^*$ is the complex conjugate of the optical transfer function $\mathbf{H}_l$.

$$\tau = \frac{S_\omega}{S_f} \quad (4)$$

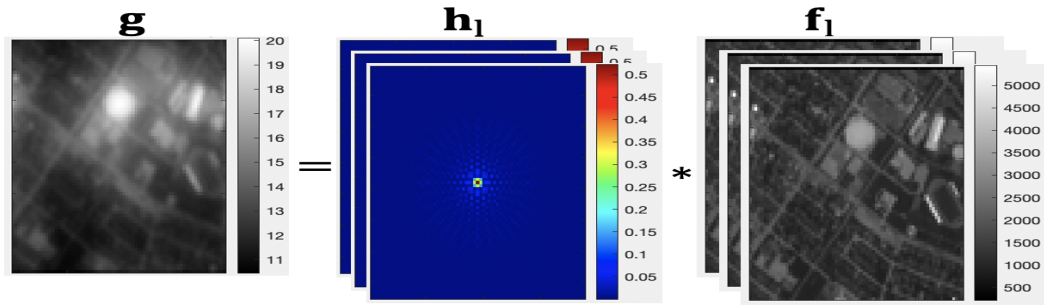


Figura 1: Sensing model

where $|\mathbf{H}_l|^2 = \mathbf{H}_l^* \mathbf{H}_l$ is the magnitude squared, S_ω is the power spectrum of the noise, and S_f is the power spectrum of the signal. The complex conjugate of complex number is $C^* = R - iI$.

3. Exercises

- To implement color computational photography system in python using diffractive elements and Bayer filter. To model the diffractive optical element use the cubic phase cubic phase, and also a set of the Zernike modes. Zernikes
- For the homework 4 use the Mc master dataset with 18 RGB images, Mc master dataset, ver Figura 2.
- To implement from scratch both the Wiener Filter and the Richardson-Lucy algorithm and compare it with the python implementation that use the skimage.restoration library.
- Plot spatial and spectral image quality metrics.
- To evaluate spatial image quality using Peak-Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM) and spectral accuracy with Spectral Angle Mapper (SAM).

La proporción de señal a ruido o Peak Signal-to-Noise Ratio (PSNR) en decibelios (dB) está dado:

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{\text{MAX}_I^2}{\text{MSE}} \right), \quad (5)$$

donde MAX_I representa el maximo valor posible para el pixel de la imagen (e.g., 255 para una imagen de 8-bit). El error cuadrático medio está dado por:

$$\text{MSE} = \frac{1}{m \cdot n} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} [I(i, j) - K(i, j)]^2, \quad (6)$$



Figura 2: Mc master RGB dataset has 18 images.

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}, \quad (7)$$

donde:

- μ_x y μ_y son las medias locales (promedios de luminancia) de x e y .
- σ_x^2 y σ_y^2 son las varianzas locales (contraste) de x e y .
- σ_{xy} es la covarianza local (similitud estructural) de x e y .
- $C_1 = (K_1L)^2$ y $C_2 = (K_2L)^2$ son constantes estabilizadoras.
- L es el rango dinámico de los valores de los píxeles (típicamente $2^B - 1$, donde B son bits por píxel).
- $K_1 \ll 1$ y $K_2 \ll 1$ son pequeñas constantes escalares.

$$\theta = \arccos \left(\frac{\sum_{i=1}^n T_i R_i}{\sqrt{\sum_{i=1}^n T_i^2} \sqrt{\sum_{i=1}^n R_i^2}} \right), \quad (8)$$

donde:

- θ es el ángulo espectral (generalmente medido en radianes).
- n representa el número total de bandas espectrales.
- T_i es el valor del espectro de prueba/píxel en la banda i .
- R_i es el valor del espectro de referencia/componente extremo en la banda i .

PSNR SSIM SAM